# Learning Whole-Body Inverse Kinematics with Obstacle Avoidance in Tendon-Driven Continuum Robots

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***Abstract*—Tendon-driven continuum robots offer inherent compliance and adaptability for operating in cluttered environments, yet their control remains challenging due to the non-rigid nature of the physical complexity that becomes inherent with the use of cables and gravitational loading. This creates a divergence between the real behavior and what the geometric models predict. This work reviews modeling and control strategies for tendon-driven continuum robots and proposes a learning-based whole-body inverse kinematics framework for obstacle-aware motion.**

## I. Literature Review

### A. Introduction

Continuum robots represent a departure from the rigid-link paradigm that has dominated industrial robots for decades. Unlike conventional manipulators, composed of discrete joints connected by stiff links, continuum robots achieve motion through the continuous deformation of flexible structures, producing smooth, organic curves similar to biological appendages such as elephant trunks and octopus tentacles [1]. This kind of compliance offers advantages in unstructured environments such as access to confined spaces and to conform around obstacles that some rigid systems might not be able to reach [2]. However this same ability to deform makes them difficult to model, control, and plan motion for.

### B. Tendon-Driven Design for Continuum Robots

Specifically, tendon-driven designs occupy a particularly important niche. These designs route cables through a flexible backbone and are anchored at distal points such that shortening the tendons would create controlled bending [3]. This tendon routing strategy allows for compact and remote transmission of actuation forces, compared to fluidic actuation systems such as pneumatic or hydraulic actuation [2].

Despite their mechanical simplicity, TDCRs exhibit complex and highly non-linear behavior. Tendon elasticity can introduce stretch and backlash while in a multi-segment TDCR , the cables controlling the distal segments must pass through all bordering segments. This leads to the creation of parasitic forces, making the tensioned cable exert unintended bending moments on every segment it passes through. These phenomena are difficult to capture using simplified geometric assumptions and can lead to significant discrepancies between the predicted model and the actual behavior of the robot, mostly at high curvature values or under gravitational loading [4].

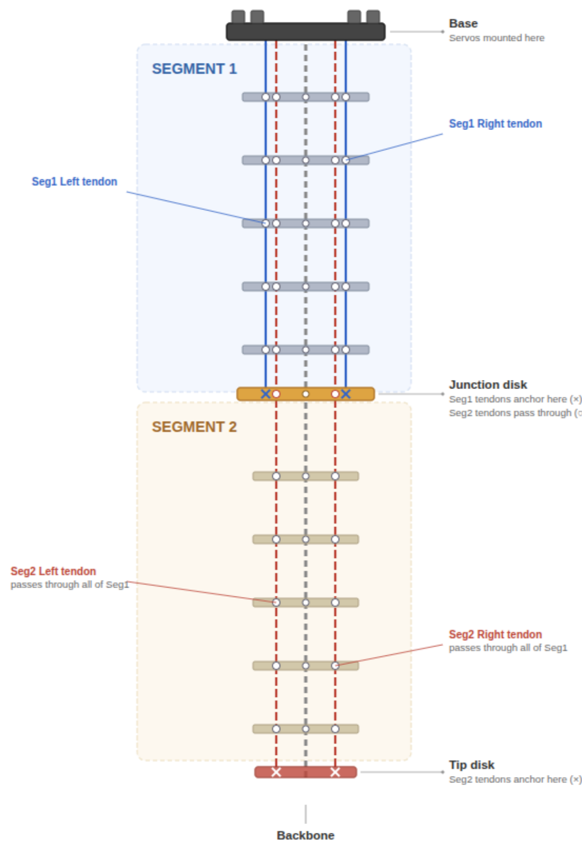



Fig. 1: Example of a 2D Planar TDCR with 2 Segments

### C. Geometric Kinematic Models

A widely adopted modeling approach for continuum robots is the piecewise constant curvature (PCC) assumption, comprehensively reviewed by webster et al. [1]. According to this framework, each independently actuated segment is approximated as a circular arc characterized by its curvature, bending plane orientation and arc length. This model has been quite popular as it helps defining simple closed-form kinematic solutions for planning and control in many continuum robot systems. The main limitation with this assumption is that it neglects distributed forces and configuration-dependent transmission effects. In tendon-driven systems, gravity, friction and tension losses can produce non-constant curvature deformations that basically violate the core assumption. As the number of segments increases, modeling errors accumulate along the kinematic chain that lead to larger deviations between predicted and actual shapes. While PCC remains valuable for low-fidelity and preliminary designs, its limitations can become an issue with cases where accurate full-body prediction is required.

### D. Mechanics-Based Models

To improve modeling fidelity, mechanics-based approaches were researched, where the continuum backbone is treated as an elastic rod subjected to distributed forces from tendons, gravity, and contact [5]. Subsequent work extended the rod-based formulations to soft continuum manipulators and showcased that these models can capture effects such as gravitational sagging, inter-segment coupling, and non-linear deformations that are difficult to represent with PCC. [6]. However, the increased fidelity comes with higher computational cost, particularly for multi-segment robots where the boundary value problems can become stiff [7]. This computational burden can limit the direct use of these models in real-time control and optimization, where many evaluations are required. These trade-offs between model fidelity and computational cost motivate the use of data-driven learned surrogate models.

### E. Learning-Based Modelling and Control

The difficulty of deriving accurate analytical models for soft and continuum robots motivated a growing body of work into learning-based solutions. One of the earliest studies by [8] showcased a feedforward neural network combined with a Jacobian-based method to solve the inverse statics of a cable-driven soft arm exhibiting non-constant curvature. The neural network was trained to approximate the nonlinear mapping between cable tensions and end-effector position, demonstrating improved accuracy compared to constant curvature models.

Thuruthel et al. [9] extended this learning-based direction by developing a closed-loop kinematic controller for a continuum manipulator operating in unstructured environments. Instead of learning a single global analytical mapping, their method learned a local differential inverse kinematics relationship that is iteratively re-evaluated using real-time feedback, enabling continuous correction of end-effector tracking errors. This work was notable for demonstrating that learned closed-loop controllers can remain effective under significant modeling errors and external disturbances, including interaction with unknown objects, where purely geometric-based kinematic solutions may become unreliable.

Although prior learning-based controllers have demonstrated trajectory execution in the presence of obstacles through end-effector feedback. Obstacle handling in these cases typically emerges from the robot's compliance and redundancy, rather than from explicitly modeling and constraining the full backbone geometry during motion. This motivates approaches that move beyond tip-centric evaluation towards a more whole-body tracking and obstacle-aware solution.

### F. Whole-Body Modeling and Obstacle-Aware Planning

Safe operation of continuum robots in unstructured environments requires reasoning over the robot's entire backbone geometry instead of just its end-effector. In comparison to rigid-link manipulators, where collision checking is performed over a finite set of links, continuum robots have a continuous spatial curve that moves nonlinearly with actuation and loading, making whole-body collision checking typically necessary for obstacle-aware operation.

Early approaches addressed this challenge through a model-based full-body estimation. Ataka et al. [10] proposed a solution, presenting a real-time obstacle avoidance framework that combined an Extended Kalman Filter (EKF) with artificial potential fields [11]. Their method estimated points along the manipulator backbone using tip position measurements, then applied repulsive potentials towards the obstacles while simultaneously guiding the tip towards the target. However, Ataka et al.'s approach relied on the constant curvature kinematic model for pose estimation and forward prediction. This means that the performance of the obstacle avoidance algorithm is closely tied to the accuracy and assumptions of the underlying kinematic model.

More recently, shape-aware control frameworks have begun integrating whole-body reasoning directly into optimization-based controllers. Kasaei et al. [12] proposed a shape-aware whole-body control strategy for tendon-driven continuum robots in endoluminal surgical navigation. Their approach combines a physics-informed backbone model with residual learning and a Model Predictive Path Integral (MPPI) controller to optimize tip tracking and obstacle avoidance. These results represent significant progress towards body-aware control, although they mostly target an application-specific setting and depend on the computational tractability of the underlying model.

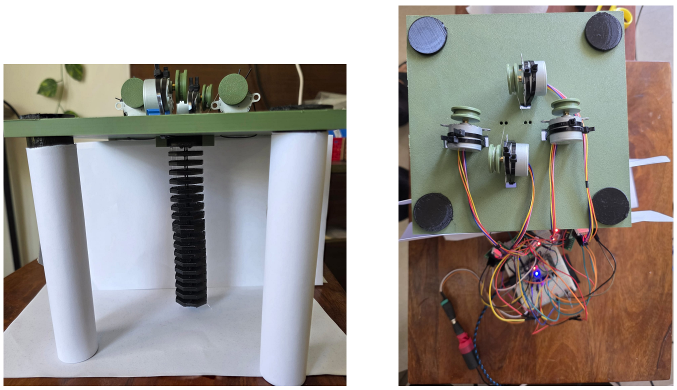
### G. Project Proposal

Building on the gaps identified above this project proposes a learning-based approach to whole-body obstacle-aware inverse kinematics for a tendon-driven continuum robot in cluttered environments. A neural network would be trained on actuation–pose data to learn the full-body kinematic mapping from tendon tensions to full-body configurations (backbone/disk positions). An optimization framework will be used to search the tendon tension space using the learned forward model as a surrogate. This is used to minimize end-effector (tip) position error with respect to a desired target, subject to the constraint that all predicted disk positions maintain a minimum clearance distance from the obstacles. As it would be predicting full backbone geometry, this constraint can be evaluated over the entire body of the robot rather than only at the tip.

## II. Experiment Results

### A. Setup

Experiments were conducted on a 2D planar tendon-driven continuum robot [2] [1] actuated by four motors. A dataset of 5000 actuation–shape pairs was collected using vision-based tracking, where the robot backbone was represented by 20 points.



(a) TDCR structure mounted on the frame

(b) Motor and control system for tendon actuation

Fig. 2: Experimental hardware setup used for data collection, showing the tendon-driven continuum robot and motor actuation system.

Unlike prior approaches that rely on simplified geometric assumptions or tip-only representations, this setup enables learning a direct actuation-to-shape mapping from real hardware data. This allows full-body behavior to be evaluated for obstacle-aware scenarios, where collision avoidance depends on the entire backbone rather than only the end-effector.

### B. Results

The following results evaluate the accuracy of different approaches in predicting robot configurations from actuation inputs, and assess their impact on control performance and collision outcomes.

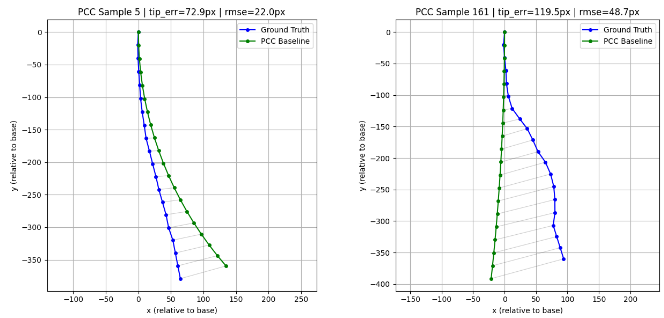
*1) PCC Baseline Results:* **Quantitative Results**

| Model | Mean Tip Error (px) | Shape RMSE (px) | PCC |
|---|---|---|---|
| PCC Baseline | 106.14 | 38.60 | 0.969 |

TABLE I: Performance of the PCC baseline in predicting robot shape and tip position.

The PCC baseline models the robot as a single planar circular arc, where curvature is computed from normalized differential tendon actuation. The robot length is estimated from tracked data, and predicted shapes are aligned to a common base frame for comparison.

**Qualitative Results**




(a) Case 1

(b) Case 2

Fig. 3: PCC baseline predictions compared to ground truth.

**Discussion** The PCC baseline captures the overall bending trend of the robot, as reflected by the high correlation (0.969). However, significant geometric errors remain, with a mean tip error of 106 px and shape RMSE of 38.6 px.

Across both examples [I], the model approximates the global curvature but deviates from the true backbone shape, particularly towards the tip. This indicates that while the constant curvature assumption is sufficient to capture general motion, it fails to model the non-uniform deformation observed in the real robot.

*2) Tip-only Neural Network Results:* A multi-layer perceptron (MLP) is trained to map tendon actuation inputs directly to the 2D tip position. Inputs are normalized spool positions, and outputs are scaled tip coordinates. The model is trained using mean squared error loss.

**Quantitative Results**

| Model | Mean Tip Error (px) | Std (px) |
|---|---|---|
| Tip-Only NN | 13.037 | 13.437 |

TABLE II: Tip prediction performance of the tip-only neural network model.

**Qualitative Results**

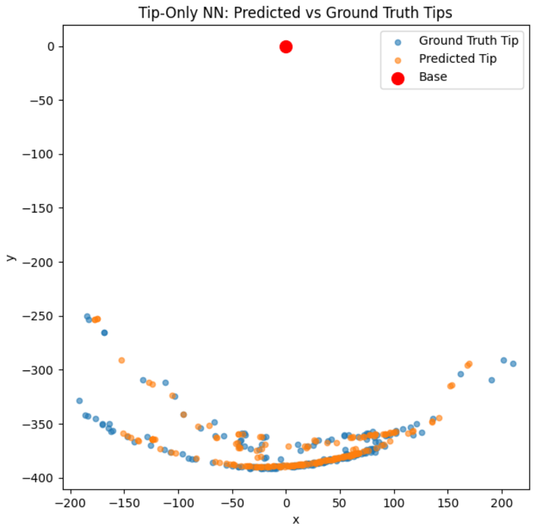



Fig. 4: Predicted vs ground truth tip positions using the Tip-Only neural network.

The Tip-Only neural network significantly improves tip prediction accuracy compared to the PCC baseline, achieving a mean error of 13 px. As shown in Fig. II, the predicted tip positions closely match the ground truth distribution.

However, this approach only predicts the tip position and does not model the full backbone shape of the robot. As a result, while the end-effector position is accurate, the intermediate configuration of the robot remains unknown.

This limitation is critical in applications such as obstacle avoidance, where the entire robot body must be considered. Therefore, although the Tip-Only model performs well for end-point prediction, it is insufficient for understanding full-body motion.

*3) Full-Body Shape Prediction with Obstacle-Awareness:* The trained neural network predicts the full backbone shape of the TDCR from tendon actuation inputs. This predicted shape is used within a sampling-based inverse kinematics framework, where actuation inputs are optimized to minimize tip error while enforcing obstacle avoidance constraints through clearance-based penalties.

**Shape Prediction Performance**

| **Metric** | **Value** |
|---|---|
| Mean Tip Error (px) | 16.17 |
| Mean Shape RMSE (px) | 6.76 |
| Mean PCC | 0.998 |

TABLE III: Full-body neural network performance on shape prediction under base-aligned evaluation.

The full-body neural network achieves low shape reconstruction error with a mean RMSE of 6.76 px and high correlation (PCC = 0.998) III. Errors increase along the backbone, with the largest deviation observed at the tip (16.17 px), reflecting accumulated uncertainty in distal segments.

**Obstacle Avoidance**

| **Scenario** | **Tip Error (px)** | **Min Clearance (px)** | **Collision** |
|---|---|---|---|
| Single Obstacle | 0.11 | 11.71 | No |
| Two Obstacles | 13.17 | 5.01 | No |

TABLE IV: Control performance under different obstacle scenarios, showing accuracy, clearance, and collision outcomes.

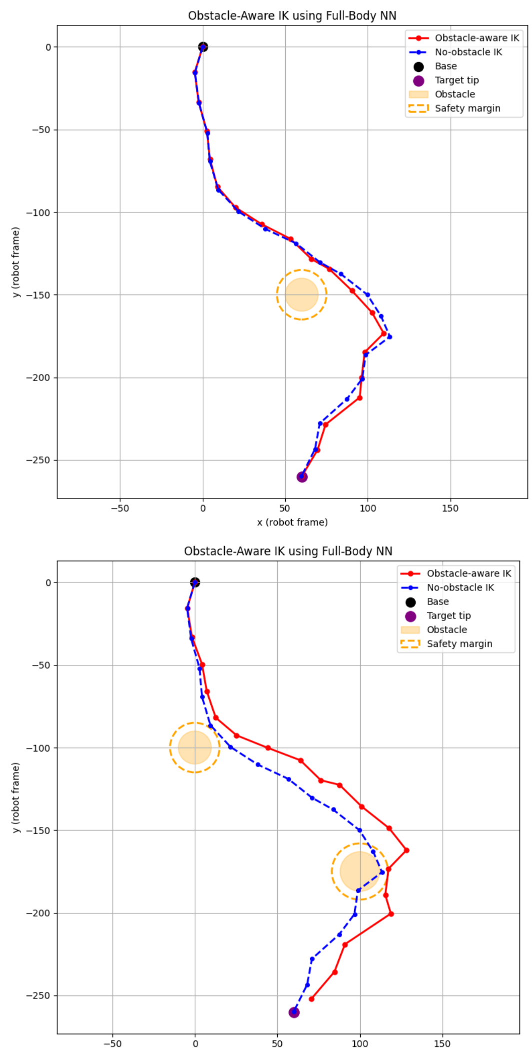



Fig. 5: Comparison of obstacle-aware and no-obstacle inverse kinematics. Left: single obstacle scenario where both methods achieve collision-free motion. Right: multi-obstacle scenario where the no-obstacle baseline collides, while the proposed method successfully avoids obstacles by adjusting the backbone shape.

In the single obstacle scenario, both the baseline and obstacle-aware methods achieve collision-free solutions, as the obstacle does not significantly constrain the feasible workspace. The obstacle-aware method maintains safe clearance without sacrificing accuracy.

In contrast, the multi-obstacle scenario introduces a more constrained environment, where the baseline solution results in collision (minimum clearance of -3.60 px). The proposed method successfully adjusts the robot configuration to maintain the required safety margin (5 px), avoiding all obstacles.

This is achieved at the cost of increased tip error (13.17 px), highlighting a trade-off between accuracy and safety. Importantly, this behavior demonstrates that full-body shape prediction enables the system to reason about collisions along the entire robot, rather than only at the tip.

## III. Conclusion

This work demonstrated that a simple learning-based approach can be effectively integrated with inverse kinematics to enable obstacle-aware control of a tendon-driven continuum robot. A lightweight neural network was used to predict the full-body shape from actuation inputs, allowing the system to reason about collisions along the entire robot.

While unconstrained inverse kinematics achieved strong tip accuracy, it often resulted in collisions due to the lack of full-body awareness. Incorporating obstacle avoidance introduces a trade-off, reducing tip accuracy in more constrained environments, but ensures safe operation.

It is important to note that the hardware operates in an open-loop manner, without feedback, which contributes to residual positioning errors and limits overall accuracy.

Overall, the results highlight that full-body shape prediction is essential for safe motion in cluttered spaces. Furthermore, the approach shows that even simple data-driven models can enable practical, shape-aware control of continuum robots without relying on complex analytical formulations.

This highlights the necessity of full-body reasoning for safe deployment of continuum robots in real-world, cluttered environments.

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